

Title

University of Glasgow



University
of Glasgow

Joseph Henry Briggs

Day Month Year

Contents

1	Balanced Homodyne for LIGO A+	4
1.1	BHD versus DC readout for A+	6
1.2	Local Oscillator Path Length Stability Requirement	11
1.2.1	Local Oscillator Path	12
1.2.2	The Linear Response Function Between Differential Strain and Power at the AS Port	13
1.2.3	Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement	15
1.3	Design of the HAM6 layout	19
1.3.1	The Constraints and Requirements of the Layout of HAM6 . .	22
1.3.2	Chosen Layout	26
1.3.3	Summary of the Chosen HAM6 layout	33
1.4	WS visulations for Analysing the Robustness of the HAM6 Layout to Mode-Mismatch	33
1.5	Conclusion about BHD for A+	42
A	Sensing and Control in the HAM6 Layout	44
A.0.1	Longitudinal Control	44

A.0.2 Alignment Control	46
B Quantum Treatment for Balanced Homodyne Detection	48

Chapter 1

Balanced Homodyne for LIGO A+

This chapter will explore the advantages, requirements, and practical design considerations for the implementation of balanced homodyne detection (BHD) for A+. In Section 1.1, BHD will be compared to the current readout scheme. Section 1.2 will derive the phase noise requirement for the local oscillator. Section 1.3 is a discussion of the functionality needed from the BHD layout, and a design that meets these requirements is shown. Section 1.4 discusses the mode matching flexibility of the chosen layout for HAM6. Section 1.5 concludes this chapter.

BHD is a method of measuring information encoded by a light field, and it forms a key part of many quantum optics experiments. By combining a signal beam with a much stronger local oscillation (LO) beam, phase modulations in the signal beam become amplitude modulations in the combined LO-signal beam. BHD allows for the quadrature of the signal beam that is to be measured to be selected, which is required for more complex topologies of interferometer.

All readout schemes for gravitational wave detectors rely on a LO field of some kind. The first generation of detector used heterodyne readout. Heterodyne readout involves a modulation-demodulation scheme whereby the signal is mixed with a RF local oscillator to a frequency where technical noise is no longer a concern, and then is demodulated at the same modulation frequency to recover the original signal. This method comes with the draw back that the quantum shot noise is enhanced.

BHD is a special case of heterodyne detection. The local oscillator is an optical field, and it has the same frequency as the signal carrying field. BHD can be distilled to four main components: the signal beam, the local oscillator beam, a balanced beam splitter, and two photo-diodes. The signal beam carries the information that is to be measured. The local oscillator beam is of higher power and acts as a stable reference frequency. These two beams are injected into different ports of a balanced beam splitter, and the remaining ports have photo-diodes placed at them. The information encoded by frequency modulations in the signal beam can be obtained through subtracting the two photo-diode signals. Due to the balanced nature of the beam splitter, the classical intensity noise of the local oscillator is not present in the BHD signal. These properties of BHD make it ideal for measuring quantum states of light. For instance, if one was trying to measure a squeezed state, by selecting the phase between the signal and the LO beams, BHD can be used to measure the phase and amplitude quadratures of the light field.

The current generation of detectors use DC readout. DC readout is actually a form a BHD. The local oscillator is an optical field generated by an offset in the arm lengths of the interferometer. This readout scheme has advantages over a traditional

BHD scheme as it simplifies the layout of the interferometer. The local oscillator and signal beams are always co-aligned, and there is no phase noise between the LO and the signal. To some extent, the amplitude and the homodyne phase can be controlled in DC readout, but it is far less flexible than the typical BHD set-up.

1.1 BHD versus DC readout for A+

There are some advantages for BHD when it is compared to DC readout, although for A+ the quantifiable difference is hard to judge. The main attraction for implementing BHD is the ability to reduce the dark fringe offset. This offset gives rise to most of the limitations associated with DC readout, such as the large coupling between the signal recycling cavity length and the DARM signal. BHD also provides the ability to select the readout quadrature, which is a requirement for novel interferometer configurations.

The current detection scheme is known as DC readout. DC readout produces a field due to a constant differential offset in the arm lengths. The field due to the differential offset is the local oscillator in DC readout. The local oscillator beats with a field created by a gravitational wave in order to produce a measurable signal. The size of the local oscillator field can be controlled with the DC offset, and the homodyne phase can be controlled with a difference in the power between the arm. This method of readout has some key advantages, such as there being no phase noise between the local oscillator and the signal carrying field, and the local oscillator and signal being perfectly aligned and mode matched. The problems with this readout technique are associated with requiring the DC offset as this light gives rise to noise

couplings.

In order for BHD to be worthwhile, it must be at least as good as DC readout. The BHD system will need to provide sufficiently low phase noise as to not degrade the GW signal. The control over the alignment and mode matching between the LO and the signal is also important, as this will cause light to be scattered back into the interferometer from the OMCs. The new layout for the BHD readout will have to provided these features.

BHD has some practical benefits over DC readout¹ as well as some interesting prospects in relation to the quantum noise limit of the detector. The practical benefits revolve around being able to eliminate the DC light required for DC readout to work. The primary benefit is in the reduction in SRC length noise coupling to the gravitational wave signal, and another benefit is being able to reduce the scattering from the septum window between HAM5 and HAM6. The quantum noise aspect of BHD arises due to the freedom of choice in the BHD angle.

One of the practical outcomes of implementing BHD is the potential to reduce SRC length noise coupling. Noise on the length of the SRC can couple to the gravitational wave signal via two routes. The dominant coupling will arise due to

¹One ‘advantage’ that is often talked about in relation to BHD is based on the electronic noise of the readout photo-diodes. This is not real advantage. As the BHD signal is obtained through subtracting the signals from two photodiodes, it is possible to obtain a very low electronic noise floor when designing the transimpedance amplifier by subtracting the currents produced by the photo-diodes rather than having two separate transimpedance amplifiers and subtracting the signals made by these later. This is not really an advantage, as low enough electronic noise can be obtained using other construction techniques that allow for small DC signals while still having signal shaping that gives a large enough SNR in the detection band. Having separate transimpedance amplifiers for each photodiode also has useful diagnostic advantages, such as being able to see the sum of the two photo-diode signals. Therefore, direct subtraction of the photo-diode signals should not be considered an advantage of BHD over DC readout.

the power entering the SRC, which is proportional to the DC offset. The other coupling arises due the optical spring between the SRC and the arm cavity length, whose strength depends on the DC offset.

The dominant coupling due to the power entering the SRC can be intuitively understood by considering a Fabry-Perot cavity. For small changes in the cavity length, the light transmitted by the cavity way that length noise may couple to the signal light is due to the dependence on the transmitted light by the SRC on its length and the amount of light entering the cavity. There is a relationship between the length noise and the power transmitted that scales linearly with the DC offset.

The second route that SRC length noise may couple to the the signal is by the optomechanical coupling due to an the optical springs between the SRC and the arm cavity. Due to this optical spring effect, a change in length of the SRC will cause the arm cavity length to change as well[1]. This effect is much smaller than the first effect, so the estimate for the reduction of SRC length coupling will only considering the first effect.

SRC length noise is currently too high for LIGO to meet the A+ requirements Shelia Dwyer post. To estimate the reduction in SRC length noise that BHD may provided, the amount the DC offset that is currently used should be compared to the amount that is proposed for BHD. LIGO currently uses at the AS port there is 23 mW, with 1 mW of residual light (light that cannot be removed)(shelia dwyer answer in email FW: BHD review committee request & document PREVIEW). Assume 0.85 A/W ($q_e = 99\%$). The AS power for BHD is between 1 and 5 mW. Therefore, a conservative estimate would for the factor that the SRC length noise

coupling is reduced by is 4.

Strictly speaking these are not properties of BHD readout, as scatter from the septum window and the SRC length noise could be reduced by other means, however BHD gives . For instance, the septum window could be removed and suspensions with additional stages could be used. These solutions however are not as practical as they sound, and are no more practical than implemented BHD. The septum window is key to being able to commission the interferometer. The number of times HAM5 and the other connected chambers are vented should be kept low, as each time this is done gas and water vapour enters the system (this contributes to the noise floor of the interferometer) and this will not be removed as this is a large engineering challenge. The design and install of three new suspensions is also a large design task, which ultimately gives less benefit compared to a BHD readout scheme.

[2][3],[4][5][6][7]. Different gravitational wave sources carry their information in different frequency bands, and the detector may be optimised to to reflect this. For instance, BNS signals continue to higher frequencies than BBH signals (due to them being smaller) so it may be of interest to optimise the detector around these frequencies to improve our understanding of the properties of BNS.

The current generation of gravitational wave detectors are going to be limited by a combination of radiation pressure noise and shot noise in their detection bands. These two noise sources are intrinsically linked by the Heisenberg uncertainty principle, and can both be understood as arising due to the number of photons used to probe the position of the end test masses. This limit in the noise floor is known as the standard quantum limit. To circumvent this limit the homodyne angle must be

free to choose, a functionality that is provided by BHD. In principle, DC readout gives the ability to set the homodyne angle, although in practice it is much less flexible than a BHD configuration.

A squeezed light field can be used to surpass the noise floor set by a squeezed state. The exact way in which the quantum noise is changed is controlled by the squeezing angle, and to detect these squeezed states the homodyne angle must be correctly set. In practice, the squeezing angle is set to be at the same angle as the readout quadrature. The effect that controlling the squeezing angle has on the SQL has been demonstrated for at LLO [3], however this used a diagnostic BHD set-up which was not intended for providing the low noise required to make gravitational wave detections. [Need to read up on this, https://doi.org/10.1103/PhysRevD.65.022002](https://doi.org/10.1103/PhysRevD.65.022002)

More advanced upgrades, e.g. see [6], could reduce quantum noise substantially, and they rely on the implementation of BHD. These changes to the interferometer are non-trivial, and would require significant redesign of many parts of the layout, so the implementation of BHD for A+ does not correspond to a suitable BHD system for one of these more advanced interferometer configurations. However, BHD for A+ would provide experience with implementing BHD on a large scale interferometer, which would help when considering how to implement these more novel interferometer configurations.

In summary, BHD has a couple of key advantages over DC readout, namely the reduction in SRCL noise coupling and the ability to shape the quantum noise floor of the interferometer in conjunction with other upgrades. The SCRL length coupling comes from the DC offset required to provide an LO beam for the signal generated

by a differential modulation in arm length to beat against. Most interferometer configurations that beat the SQL assume that the homodyne angle can be set. The drawbacks of BHD when compared DC readout are all engineering challenges, nothing fundamentally favours DC readout to BHD.

1.2 Local Oscillator Path Length Stability Requirement

This section will go through the steps required to derive the phase noise requirement for the beam path of the A+ LO. A semi-classical approach has been used, and the end result is the same as if a full quantum mechanical treatment was used. In fact, the quantum mechanics of this problem is included in the A+ sensitivity curve and the expected power in the signal beam. The phase noise of the local oscillator must be kept low enough so that it does not result in intensity fluctuations greater than those due to the shot noise (which can be squeezed) of the light in the signal beam.

Phase noise arises due to phase fluctuations in the LO beam being indistinguishable is phase fluctuations in the AS beam. A gravitational wave produces a phase modulated signal in the AS beam, and the LO field phase is modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be sufficiently low that the signal that is generated by the interference of the LO and AS beams corresponds to the phase modulation of the AS beam.

The relationship between the differential phase in the arms and the power in the signal beam was used to find a light noise curve for the beam exiting the SRM in A+. The linear response function due to a differential phase change in the arms was computed with FINESSE, and this was used to deconvolve the strain sensitivity curve to produce a light noise curve for the light exiting the AS port of the interferometer.

This light noise curve was then used to set the maximum allowable intensity fluctuations due to phase noise between the local oscillator and the signal beams. This phase noise can then be converted into a distance, which sets a stability requirement on the steering optics used to combine the LO and signal beams. From this, it is found that the optics need to be suspended as a triple stage pendulum.

1.2.1 Local Oscillator Path

Two options for the local oscillator path were considered: a beam picked off from the main beam splitters AR coating (BSAR) or the beam transmitted through PR2 [8] (POP). The determining factor in choice of LO path was the effect of backscatter, and the POP choice is far superior to the BSAR choice at all frequencies and scatter phases. This is because any backscatter from the BSAR enters only one arm, and therefore is a differential effect. Any back scatter that enters the POP port is common-mode, meaning that the differential arm signal is far less sensitive to this source of backscatter.

The layout for the optics which combine the the LO and AS beam is shown in figure 1.1. The beam transmitted by PR2 is used for control of the interferometer, so BHDBS1 is needed to pick off some of this light. The LO beam is steered towards

the combining beam splitter BHDBS2. with BHD1. Two lenses (**need names for these**) are used for mode matching the LO beam to the AS beam. The AS beam is transmitted through the SRM and OFI, and then is steered towards BHDBS2 with OM0.

1.2.2 The Linear Response Function Between Differential Strain and Power at the AS Port

The linear response function of the A+ detectors was obtained using FINESSE. Since the core optics of the interferometer will not be changing in a way relevant to this calculation, the LLO model[9] was used to calculate the response between differential motion of the ETMs and the power transmitted through the OMC. The effect of the light being transmitted through the OMC has no effect on the linear response function at the frequencies contained in the detection band, and so the OMC has no effect.

FINESSE allows one to measure linear response functions by applying a ‘swept sine’ signal to a component, and demodulating the light at the frequency of the applied signal. To apply a differential arm signal with FINESSE, **fsig** commands were used to apply two signals in antiphase with each other to the ETMs,

```
fsig sig1 ETMXHR 1 0
fsig sig2 ETMYHR 1 180.
```

The frequency of this modulation was swept, and the signal produced at the AS port was measured by demodulating the light at the modulation frequency. The commands

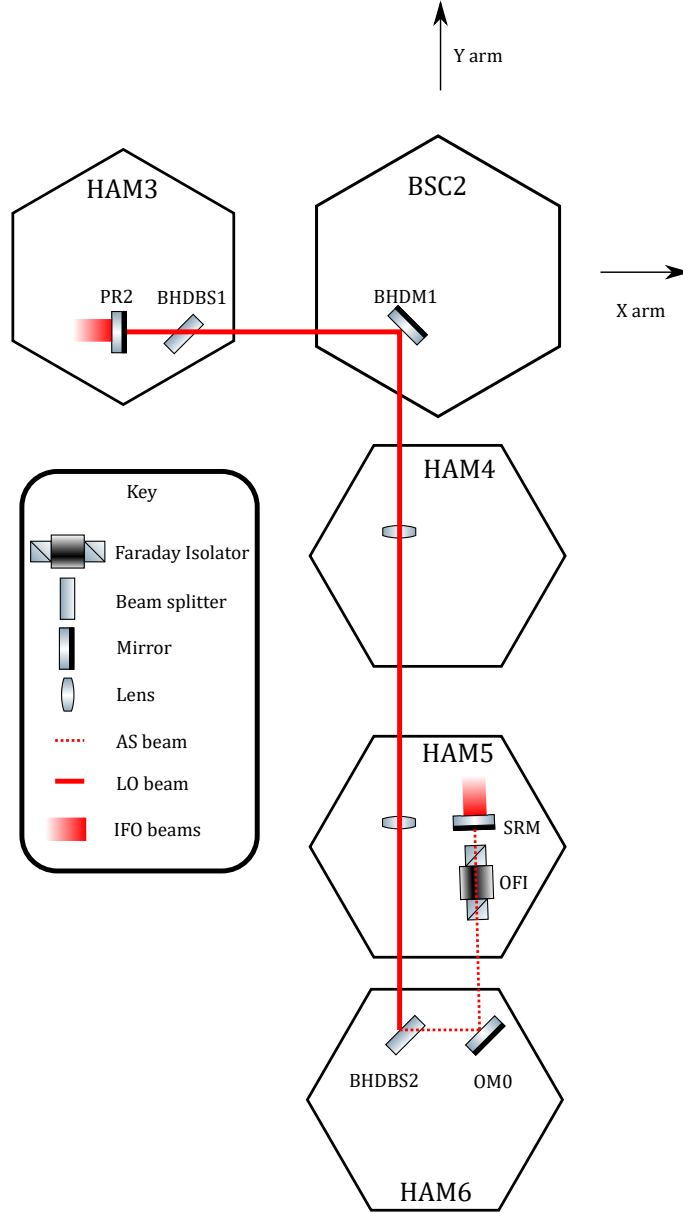


Figure 1.1: Sketch of the LO and AS beam paths. The LO beam is obtained from the light transmitted by PR2. Some of this light is needed for control of the interferometer, so BHDBS1 is used to split the light. The LO beam is then directed towards the combining beam splitter BHDBS2 with BHDM1. Two lenses are needed to mode match the LO beam to the AS beam. The AS beam is transmitted through the SRM, passes through the OFI, and is directed to BHDBS2 via OM0.

```
xaxis sig1 f log 5 5000 2999
```

```
put P_DC_OMC_probe f1 $x1
```

do this.

To get the linear response function between arm motion and AS power into units of $\frac{\text{Power}}{\text{Strain}}$ (or $\frac{\text{W}}{\text{h}}$), several factors of 2 are needed including. Since the input modulation is doubled by moving the mirrors differentially the response needs dividing by 2. To account for the ETMs retro-reflecting the light (**or is the weird Finesse convention**), the response needs dividing by 2 again. To convert this from phase units to strain units, a factor of $\frac{\lambda}{2\pi L_{\text{arm}}}$ is needed. Therefore, the output of the the FINESSE simulation needs to be multiplied by $\frac{1}{2} \frac{1}{2} \frac{2\pi L_{\text{arm}}}{\lambda}$ to be in units of $\frac{\text{W}}{\text{h}}$. The linear response function between a differential arm strain and the power at the AS port is shown in figure 1.2.

1.2.3 Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement

The sensitivity curve will be shown in an earlier chapter. To convert the A+ sensitivity curve into a path length stability requirement for the local oscillator, a few steps are required. First, the sensitivity curve must be expressed in units of $\frac{\text{W}}{\sqrt{\text{Hz}}}$. Second, the intensity noise of the AS beam needs to be converted into a phase noise. In order for the phase noise of the local oscillator to not limit the performance of the gravitational wave detector, the phase noise of the AS beam must be greater than the phase noise of the LO beam. Finally, a factor of 10 safety margin is required. The HRTS requirement is then obtained by including the number of reflective optics

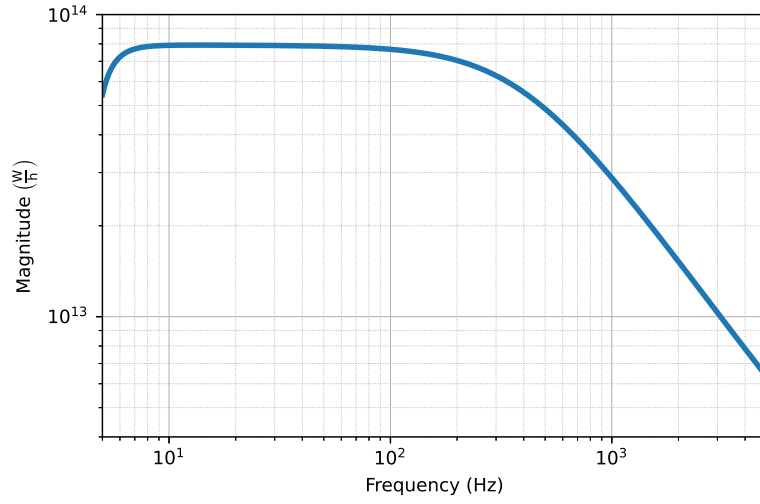


Figure 1.2: The linear response function between strain and light power at the AS port was obtained from a FINESSE simulation of the LLO detector[9]. The corner frequency at ≈ 300 Hz comes from the pole of the coupled SRC-arm cavity. The behaviour below 6 Hz is due to optical spring effects.

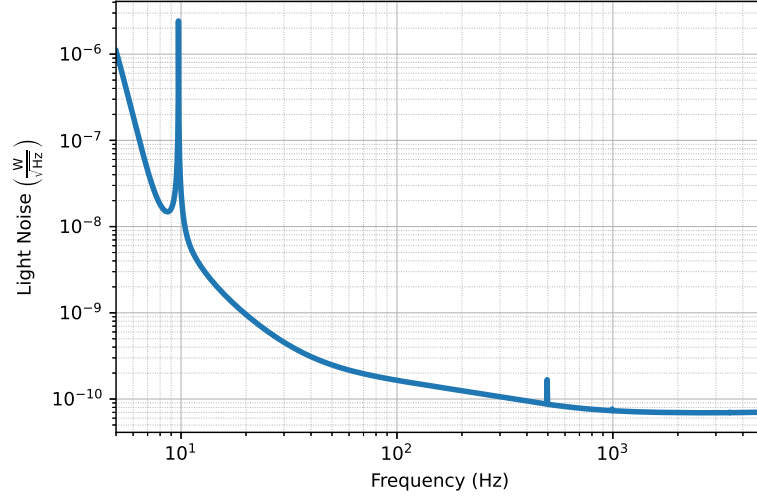


Figure 1.3: The intensity noise of the light at the AS port for the A+ detector. This was obtained by deconvolving the detectors sensitivity curve[10] with the function shown in figure 1.2.

in the LO path.

The A+ sensitivity[10] curve was multiplied by the function shown in 1.2 to convert it from units of $\frac{h}{\sqrt{\text{Hz}}}$ to $\frac{W}{\sqrt{\text{Hz}}}$. This is shown in figure 1.3. At frequencies above 1 kHz, the requirement is flat as this is the region dominated by shot noise. At lower frequencies, the requirement is relaxed as the sensitivity curve becomes dominated by other noise sources.

Next, a change in power measured in the combined LO/signal beam due to a phase modulation (i.e. path length modulation) in the LO or signal path needs calculating. When the LO and AS beams interfere at the beam splitter (BHSBS2), the power of

the resulting beam is

$$P = |E_s e^{i\omega t} + E_{\text{LO}} e^{i\omega t + \phi}|^2 = |E_s|^2 + |E_{\text{LO}}|^2 + 2|E_s E_{\text{LO}}^\dagger| \cos(\phi), \quad (1.1)$$

where E_s and E_{LO} are the (complex) field amplitude of the signal and local oscillator beams, ω is the laser frequency, t is time, and ϕ is the phase difference between the local oscillator and the signal beam, also known as the homodyne angle. Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_s E_{\text{LO}}^\dagger| \sin(\phi) d\phi. \quad (1.2)$$

and since the homodyne angle is 90° , this becomes

$$dP = -2|E_s E_{\text{LO}}^\dagger| d\phi. \quad (1.3)$$

To convert ϕ to distance, use

$$\frac{\lambda}{2\pi} d\phi = dx. \quad (1.4)$$

The light noise curve shown in figure 1.3 should be multiplied by $\frac{\lambda}{2\pi} \frac{1}{2|E_s E_{\text{sig}}^\dagger|}$ to obtain the the noise floor of the interferometer in units of LO path length noise.

To determine the longitudinal motion requirement for the optics that combine the LO and signal beam the number of optics and a factor of ten safety margin need to be included. The requirement assumes, based on the current design, that there will be 2 mirrors at 45° which are used to steer the AS and LO beams. As the optics reflected at 45° , the path length change due to the optics motion is multiplied

by $\frac{2}{\sqrt{2}} = \sqrt{2}$. The beams are combined on a beamsplitter, and the notion of this beam splitter needs to be counted twice. A sketch of this, Figure 1.4, shows why this is the case. A factor of 10 is then included as a safety margin since there are many uncorrelated noise sources in LIGO. For a 98 mW LO beam and a 5 mW AS beam, the noise floor of the interferometer in units of LO path length is shown in figure 1.5. The requirement for the longitudinal motion of optics in the LO or signal path at 100 Hz is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$.

To determine the amount of isolation required, a typical motion of the optical table needs to be compared to the longitudinal motion requirement for the optics. A typical motion of the optical benches at 10 Hz is $10 \times 10^{-1} \text{m}/\sqrt{\text{Hz}}$ [11]. To achieve this with passive isolation, a triple suspensions will be used. These suspensions are known as 'HAM Relay Triple Suspensions' (HRTS).

1.3 Design of the HAM6 layout

Fundamentally, the layout in HAM6 must combine the AS beam and the LO beam, and sense the combined beams to form a balanced homodyne detector. There are several other functionalities required and constraints imposed on the layout of HAM6. The following section will describe these functionalities and constraints and how they guided the design of the HAM6 layout. This work was undertaken by a team including myself, Ken Strain, Russel Jones, Stephen Webster and Mark Barton.

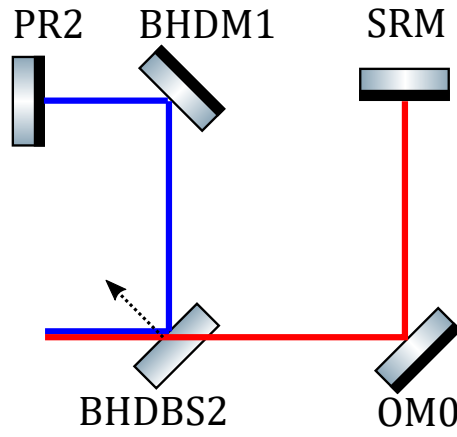


Figure 1.4: A sketch of the LO and AS beam paths. The LO beam path is shown in blue and the AS beam is shown in red. A dashed line indicates the direction that the recombining beamsplitter, BHDBS2, moves. When this happens, at the point of the beams combining the phase of the AS beam has increased while the phase of the LO beam has decreased. This means that the recombination beamsplitter motion needs to be counted twice.

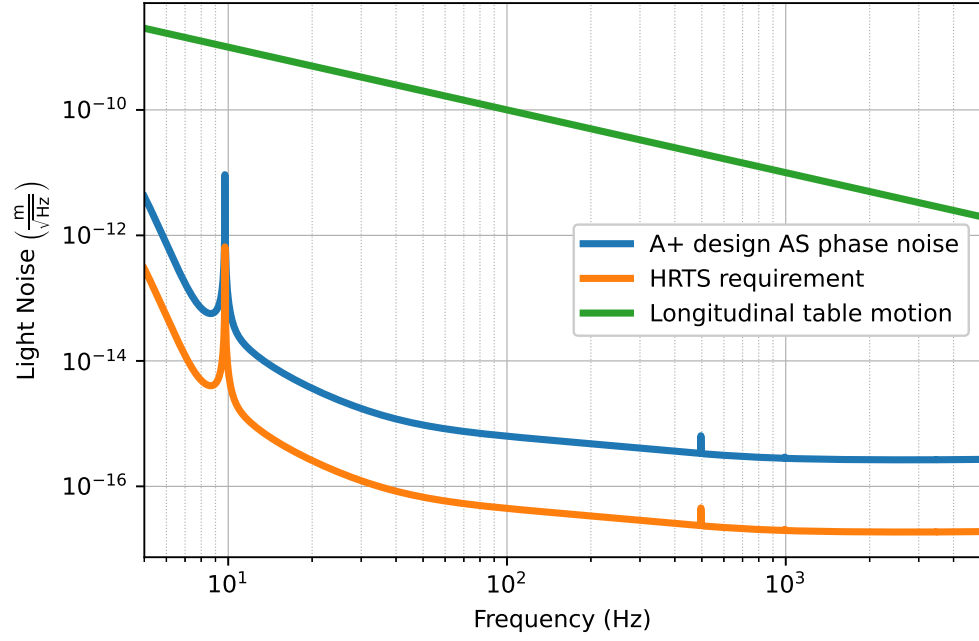


Figure 1.5: The phase noise between the LO and the AS beams causes noise. The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve is shown in units of path length noise between the AS and LO beams, and the orange line shows the requirement of HRTS mirror motion. **Replot, the y axis is shit!**

1.3.1 The Constraints and Requirements of the Layout of HAM6

This section will describe the functionality required from the the HAM6 layout, and some practical constraints that are imposed by the hardware and the infrastructure. The layout in HAM6 must provide steering and mode matching between the interferometer and the OMC. The main design challenge for this layout was to achieve 45° of Gouy phase between the mode matching optics, as this requires a lot of space. There are a few properties of the HAM6 layout that cannot be changed, such as the entry point of beams and size of the optical table.

As the constraints give a starting point for all designs, they will be discussed first². These include: where beams enter the HAM6 chamber, the size of the OMC waist, and the size of nominal mode entering HAM6. The size of the HAM optical table does restrict the design, however small overhangs sometimes are permitted as table extensions can be made.

The entry point of the AS and LO beams in principle can be anywhere in HAM6, however in reality they are fixed due to the location of HAM6 with respect to the interferometer, due to the location of the septum windows between HAM5 and HAM6, and due to the many other components of the interferometer that these beams need to avoid. The LO beam is steered with BHDM1 onto BHDBS2, and

²It could be argued that requirements are a constraint in a design. Here, the constraints are geometric properties which we have no freedom in, such as the input beam shape, the OMC geometry and the layout of the rest of the rest of the interferometer. A requirement, such as the desired minimum mode matching, is a property which doesn't have a set 'physical form'. For instance, while Gouy phase does correspond to a physical distance, not all 45° Gouy phase spaces are the same physical size.

the AS beam is steered with OM0 onto BHDBS2. This is required to provide the degrees of freedom to co-align the AS and LO beams. There is some freedom in choice for the location of OM0 and BHDBS2, but there is no choice in the location of the AS and LO beams.

The design beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i$ m, so the design of the HAM6 mode matching optics is based on this number. In reality, the beam parameter could be different to this, as discussed in chapter ??). At the SRM, there are two sets of modes we could design the AWC to match to: The arm mode, or the SRC mode. Ideally, these would be the same mode, however geometry errors in the SRC cause the arm and the SRC to be mismatched meaning the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM. This set of likely modes, shown in ??, was taken into consideration when choosing a layout for the AWC in HAM6.

The current OMC design (with trivial modifications) will be used to remove junk light from the signal beam, so there is no choice in target waist size. The input beam at the SRM is defined in T1200410. The mode matching is designed assuming that the beam at the SRM is the ‘design mode’. Mode matching between the beam exiting the SRM and the OMC waist will be done with a static lens and 2 active mirrors. The static lens will be placed just after the OFI and its purpose is to provide sufficient Gouy phase separation of the 2 active mirrors.

It was decided that the two beams after BHDBS2 would have the same path length to reduce the cost of the HAM6 optics. This may also improve the common-mode noise rejection ratio.

The requirements for the HAM6 layout are to provide steering for the interferometer beam into the OMC, and mode matching between the interferometer beam and the OMC. The goal for mode matching is 98%. After BHDBS2 the layout must have 2 mirrors separated 90° and 45° of Gouy phase to provide orthogonal control over the beam pointing and mode-matching. The 45° Gouy phase separation requirement is what makes the design of HAM6 none trivial.

The goal for mode matching between the arm mode and the OMC mode is is $> 98\%$. The SAMS actuators will provide ± 100 , mD of optical power tuning range, with a bandwidth of the order 100 Hz. Astigmatism can be considered as a loss, and none-zero angles of incidence (AOI) of beams onto curved optics results in a the beam becoming astigmatic. This prevents designs using large AOI. This is described in detail in Chapter ??.

In principle, the role of the AWC in HAM6 is to correct for geometry errors of the optics after the SRM such that the mode matching between the Arm mode and the OMC waist is maximised. This assumes that the changes in the interferometers mode, for instance due to a change in thermal state, will be corrected for by other subsystems e.g. the TCS. In practice, operators of the interferometer do what ever improves the SNR of the the gravitational wave signal. For instance, it has been found that TCS can be used to decrease laser intensity noise coupling to the the GW signal channel. The AWC in HAM6 was designed with this flexibility in mind. The losses due to components in HAM6 is summarised in table 1.1 (See E1900385).

While in principle only two curved mirrors would be required to mode match the beam from the interferometer to the OMC, in fact a single mirror may be able to do

that, a lens in the OFI will be used to provide further Gouy phase separation between the two active mode matching mirrors after BHDBS2. An active lens was considered here, but the unavailability of flint glass meant that this idea was discarded. There were also concerns about the heating of the OFI assembly that an active lens may cause. As well as this, the range of optical powers for the SAMS is large enough that it would be an over-design to implement an active lens in the OFI.

To provide orthogonal mode matching control, two of the mirrors should be separated by 45° . This Gouy phase separation requirement give us a method of generating candidate designs, as these can be interpreted as needing the mode matching mirrors to be as far apart as possible from each other on the HAM table. As the Rayleigh range of the beams is on the order of a meter, and most gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will have to travel quite far across the table before it acquires sufficient Gouy phase.

Additionally, To provide orthogonal alignment control in each beam path after BHDBS2, two of the mirrors are required to be separated by 90° of Gouy phase. This was always done with the mirror immediately after BHDBS2 and the mirror immediately before the OMC. It was found that placing the first mirror close to BHDBS2 often provided the best layouts.

Light scatter from the OMC is considered a potential source of noise. The OMCs are expected to be the largest contributor of back-scatter from HAM6 back to the interferometer. By having the OMCs after the AS and LO beams are combined, and by having the OFI protect the interferometer, the requirement on the motion of the OMCs is fairly lax. Nevertheless, a single stage suspended platform is required.

This will provide isolation down to 4 Hz.

Additional sensors in HAM6 are required for controlling various degrees of freedom within the layout. Some of these degrees of freedom will be monitored using beams which are picked off from the main beam path, and so additional mirrors will be required to steer these beams around the chamber. A set of smaller optics will be installed to provide diagnostic beams which can be imaged with CCDs. A shutter will be required to protect the hardware in HAM6 from being damaged by the large amounts of light that can potentially be transmitted through the SRM. Appendix A describes how the longitudinal and pointing degrees of freedom may be sensed and controlled.

HAM6 loss Source	Target for O5
OMC throughput	< 2%
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	< 2% (as low as possible)
Any other optical loss	< 1%

Table 1.1: Target for losses in HAM6. Reproduction of E1900385

1.3.2 Chosen Layout

The design that is being taken forward at the time of writing is known as the ‘Pentagram’³, and is shown in figure 1.6. Optically, there are many layouts which could have satisfied the above requirements and constraints. Ultimately the choice of layout for HAM6 was determined by mechanical factors such as table space and

³It was called this as when viewed from above the main beam path looks like a pentagram. An alternative and much less interesting name for this layout is ‘5f’.

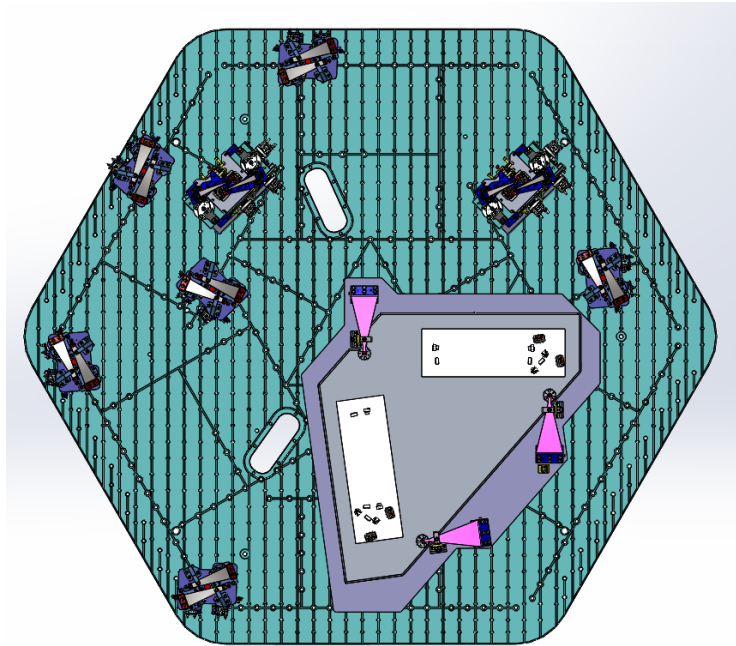


Figure 1.6: Place holder graphic. The pentagram layout. CAD will be inserted later.

number of suspensions. The optical properties of this layout are summarised in table 1.2.

The beam size and Gouy phase as the a function of position along the optical axis is shown in figure 1.7. The positions of significant points along the optical axis is also shown. The size of the beam at the final mode matching optic matches the size of the target mode, as the size of the beam cannot be changed at a mirrors surface. The beam is always less than 1.2 mm within HAM6.

The astigmatism loss was calculated to be $> 0.05\%$. This astigmatism loss was plotted as a function of angle, and above 10° the loss becomes significant. The compromise that is made when choosing a larger AOI is the ability to have more Gouy phase separation between OMx1 and OMx2. In reality, the IFO and OMC will also have a small amount astigmatism. Losses of $> 0.05\%$ are considered unimportant, especially when compared to the other expected losses, some of which are expected to be at the many percent level, in the readout of the interferometer.

The focal lengths of the OFI lens, OMx1 and OMx2 where optimised so that there was a close to 45° Gouy phase separation between OMx1 and OMx2 as possible. This is shown in figure 1.8. With this layout, it was possible to achieve 43° Gouy phase separation between OMx1 and OMx2. Other layouts were able to provide exactly 45° Gouy phase separation, however the Pentagon layout offers many practical benefits over the other considered designs that outweigh an additional 2° of Gouy phase separation. This layout also provides good Gouy phase separation for alignment control, with there being 88° between OMxS and OMx2.

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.0583	-17.1552	-
OMx1	-0.3539	-2.8290	5.6580 (concave)
OMx2	-0.8737	-1.1445	2.2891 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	432
OM0 to BHDBS2	905
BHDBS2 to OMxS	225
OMxS to OMx1	1240
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	$485 \text{ }\mu\text{m}$
OMx1 to OMx2 Gouy phase	43°
OMxS to OMx2 Gouy phase	88°
OMxS and OMx2 AOI	10°

Table 1.2: Mechanical and optical properties of the Pentagon layout.

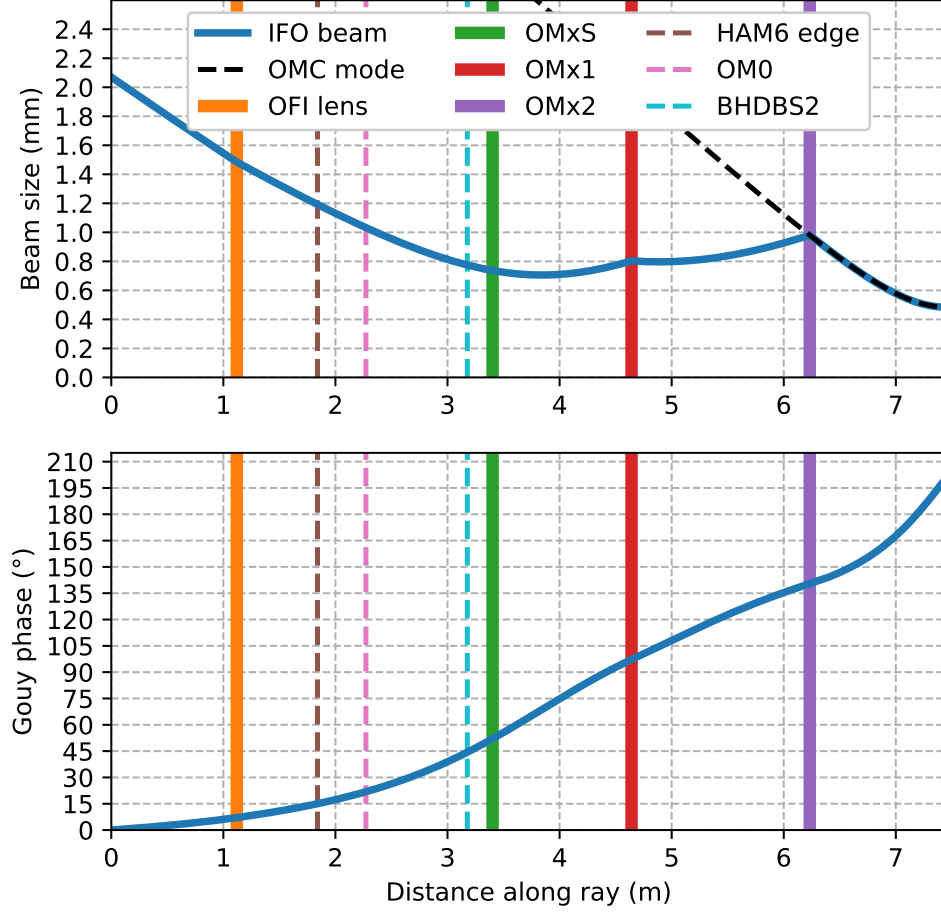


Figure 1.7: Top panel: the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Bottom panel: The gouy phase shift accumulated by the beam as it propagates along the optical axis.

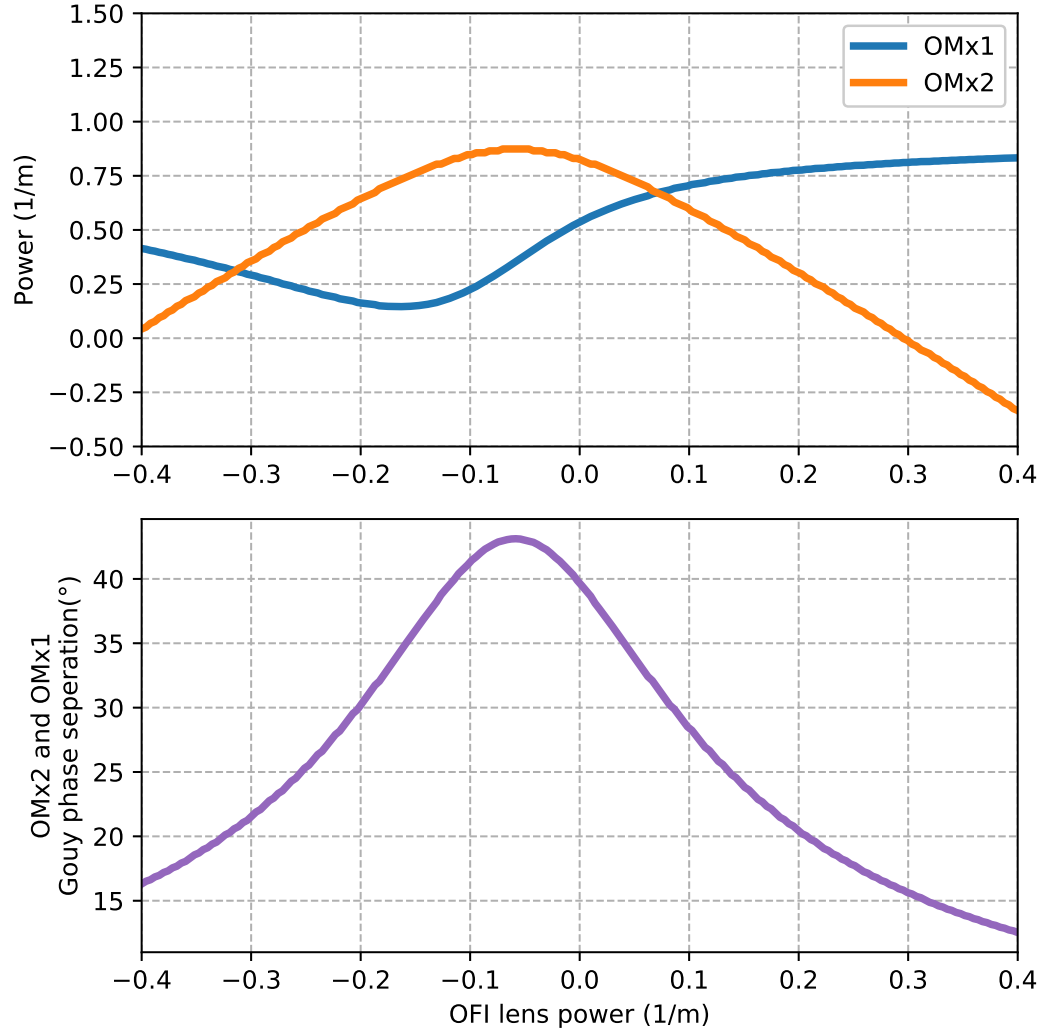


Figure 1.8: OFI lens power effects the optimal combination of OMx1 and OMx2 power required for 100% mode matching. The best choice of these lenses is the pair which gives the Gouy phase separation closest to 45° .

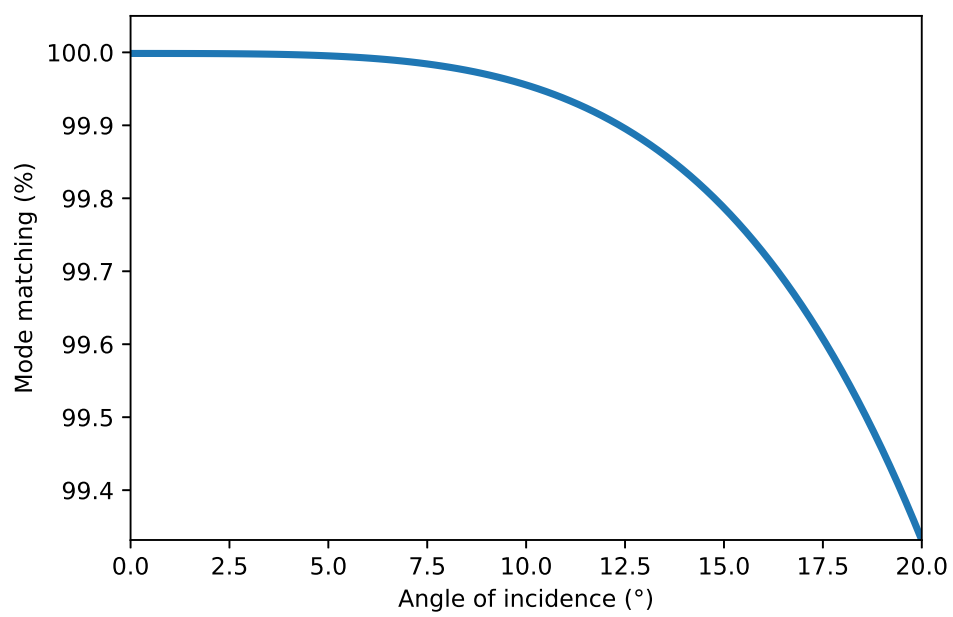


Figure 1.9: Mode matching degradation as a function of angle of incidence for OMx1 and OMx2.

1.3.3 Summary of the Chosen HAM6 layout

In conclusion, a HAM6 layout that fulfils the criteria set in Section 1.3.1 has been designed. This layout is shown in Figure 1.6. Near optimal mode-matching control and alignment control was achieved, and all the other functionality required from the layout have been considered. The next stage of development for this layout involves the design and prototyping of parts, and the consideration of fine details such as cable routing.

1.4 WS visulations for Analysing the Robustness of the HAM6 Layout to Mode-Mismatch

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, the two modes can be mismatched by being misaligned or by having a different beam shape. Mode matching typically refers to the overlap of the two modes if they were perfectly aligned.

Mode-matching is usually, to first approximation, thought of as a difference in beam waist size and beam size location. This can be mathematically described by the $HG_{20} + HG_{02}$ mode. The effect of changing the waist location of the mode is equivalent to an in-phase change in HG_{20} content, whereas an change in waist location is orthogonal to this. [read up on this](#). Sensing the $HG_{20} + HG_{02}$ mode therefore gives a measure of the mode-matching between two lasers. As the $HG_{20} + HG_{02}$ acquires 3ϕ compared to a plane wave, and 2ϕ compared to the HG_{00} mode,

spacing the actuators at 45° will give orthogonal control of the mode matching. This means that the beam size and beam waist can be independently controlled.

Two perfectly mode matched beams will have identical beam parameters. If the input mode and the target mode differ in beam size and defocus, then some light will be scattered into the $\text{HG}_{20} + \text{rmHG}_{02}$ mode. The amount of mode matching $\mathcal{M}$ between two astigmatic modes, $q_{1,2}$, is

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (1.5)$$

For a stigmatic beam set $q_x = q_y$. The mode matching can be calculated at any point along the optical axis of the layout, as the mode matching must be preserved, but it makes sense to choose a physically significant plane such as the input plane, a plane containing a curved optic, or the plane containing the waist of the target beam.

A curved mirror can only change the curvature of a Gaussian beam, meaning that at the mirrors surface the beams before and after reflection must have the same size. The process of a Gaussian beam described by q_1 being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i \frac{\lambda}{\pi w_1^2}. \quad (1.6)$$

To visualise how a mode is changed by an optical layout, beam parameters can be represented in a 'WS space'. This space parametrises a beam in terms of its size, w , and its defocus, S . This has the advantage of showing the beam in terms of

practical quantities. The downside to this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. As q is a complex number, the beam parameter could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and s produces plots where the effect of a focussing elements and propagation is easy to visualise, as focussing is just a change in defocus and propagation follows a circular path. However, this technique obscures the beam size, and as these graphs are being used to visualise the beam, rather than design the optical layout⁴, the WS representation was chosen.

To demonstrate how orthogonal actuators can be represented in WS space, consider the two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will orthogonally controllable. In a WS space, the orthogonality of two actuators can be seen by the angle between the two vectors that represent the actuation. If the two lenses are orthogonal, then the actuation vectors will be orthogonal. Figure 1.10 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase. If there is large amounts of astigmatism, it no longer makes sense to represent the beam with just one WS space diagram.

⁴This representation is useful when designing an optical layout by hand, something which is not often done anymore.

The mode matching requirement for LIGO is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode, the mode matching between two arbitrary modes can not be deduced from these contours. An example of a situation where this is obvious is two different modes may be, say, 90% mode matched to the target. As they are different from each other, they cannot be 100% mode matched.

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS from which the target mode matching space achieved was computed. To do this, first the ring of modes in WS space that is 98% matched after reflection from the second active mirror, OMx2, was considered. These modes must be within $S_{\text{OMx2}} \pm \Delta S$, where S_{OMx2} is the static power of OMx2 and ΔS is the actuation range of the SAMS. If OMx2 had no actuation, then the effect of OMx2 would be to move the ring vertically in the WS plane. However, as power of OMx2 can be increased or decreased, the shape of the ring of modes that can be mode matched to 98% can be changed. Crudely, the shape of this space can be seen by applying $S_{\text{OMx2}} \pm \Delta S$ to the 98% ring, producing 2 new rings. If the ring is cut by a line which intersects the locations on the ring at the extrema of the beam size, then two regions of the 98% ring can be created. For one of these regions, to maximise the amount of WS space covered by the actuator, the actuation applied should be $+\Delta S$, whereas for the other region, the actuation should be $-\Delta S$. This gives a set of modes which can be propagated to the first mirror, OMx1, where the same procedure can be applied. This process shown in Figure 1.11.

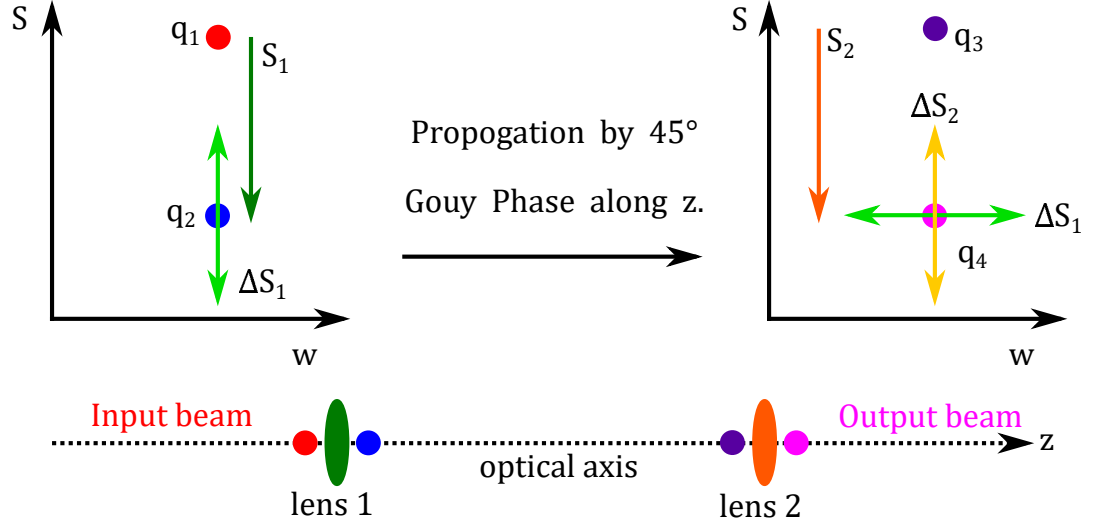


Figure 1.10: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} + \Delta S_{1,2}$. The left hand axis shows the WS plane at lens 1. The input beam q_1 is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 , which is shown as a blue dot. A light green arrow indicates the actuation of lens 1. The right hand axis shows the WS plane at lens 2. The input mode at lens 2, q_3 , is shown as a purple dot, and the output mode q_4 is shown as a pink dot. The static and dynamic power of lens 2 is indicated by the orange and yellow lines. As lens 1 and lens 2 are separated by 45° of Gouy Phase, the actuation provided by lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by lens 2, provided that the actuations are small enough.

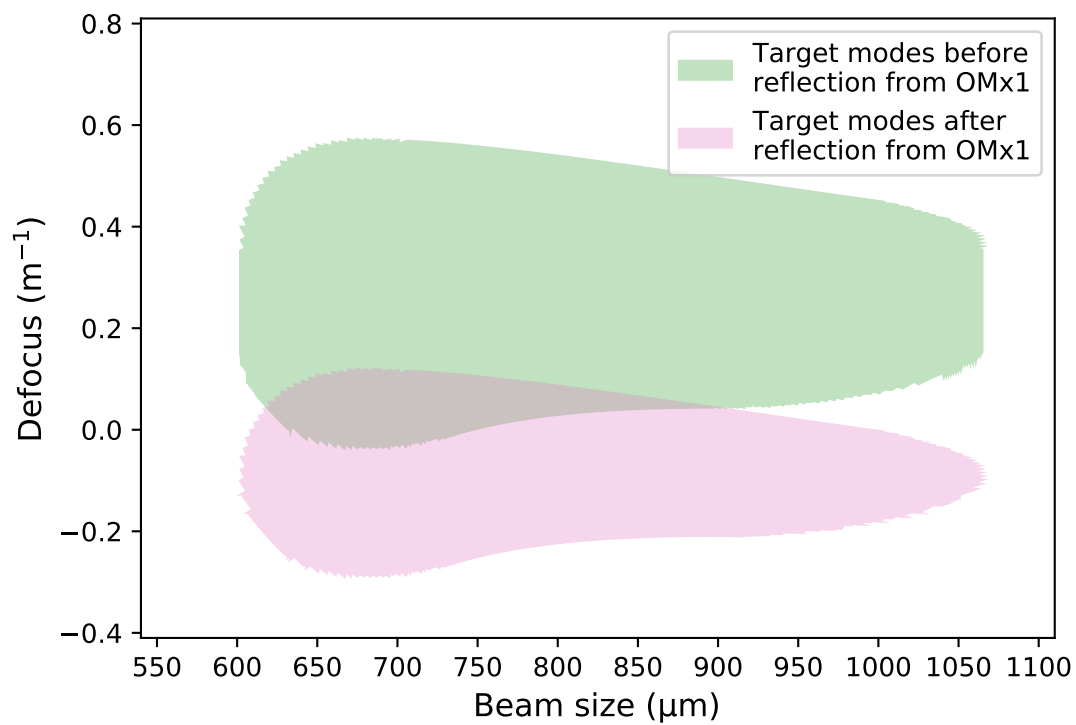
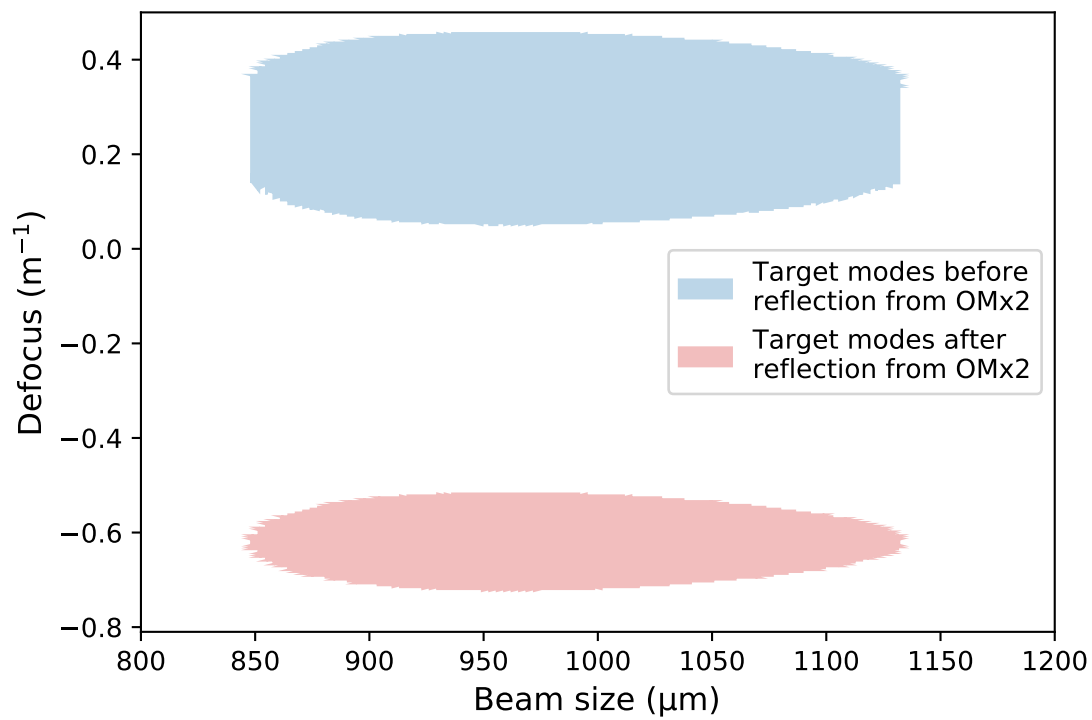


Figure 1.11 (*previous page*): Top panel: This is the WS space at the OMx2 plane. After reflection from OMx2, a mode must be at least 98% mode matched to the waist as there are no further active mode matching elements in the layout of HAM6. The target modes are shown by the red region. The blue region represents which modes can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, which approximately corresponds to modes with a defocus greater than 0.6 or less than 0.6. The line that splits the red area into an upper and lower region corresponds to the line that connects the two extremes in width of the region, and does not have to be a horizontal line. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them. These regions were then joined by a vertical line that connects them.

Bottom panel: This is the WS space at the OMx1 plane. The pink mode represent the target modes that a mode must be able to reach by being reflected from OMx1. The blue region from the upper panel was propagated back to OMx1 to form the pink region. In a similar way as before, the pink region was split into an upper and lower region, and the green region was constructed from this. Modes in the green region are capable of being transformed to the pink region, which therefore means they can be transformed to the red region, meaning they satisfy the 98% mode matching requirement. The green region was propagated back to the through the layout to SRM, and this is as the purple region in Figure 1.12.

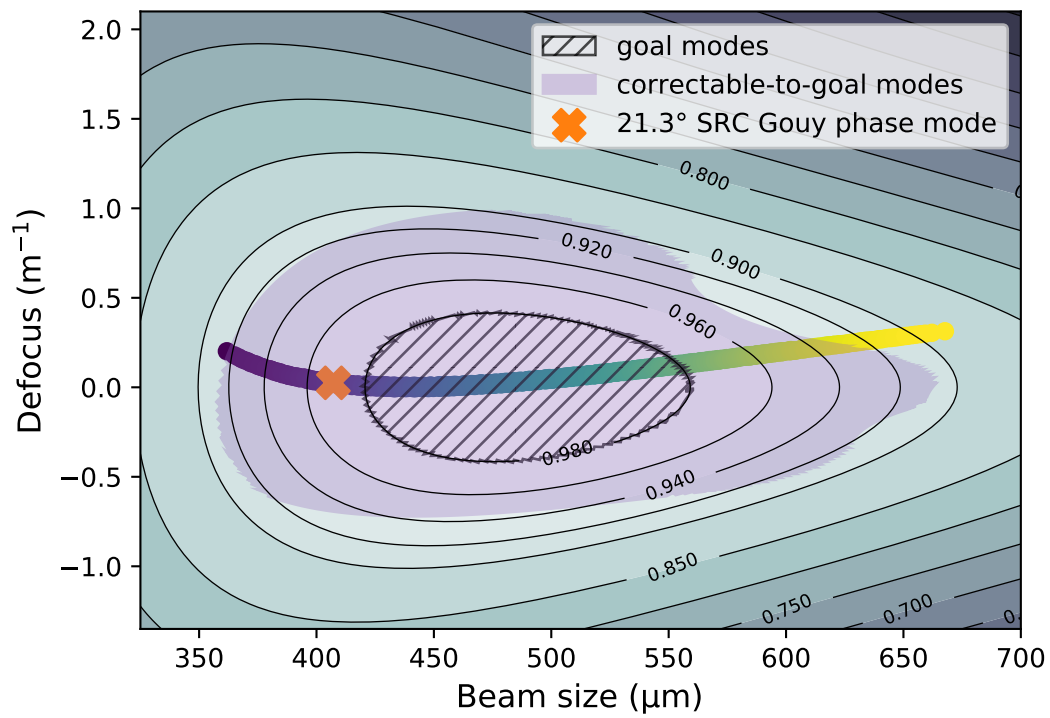
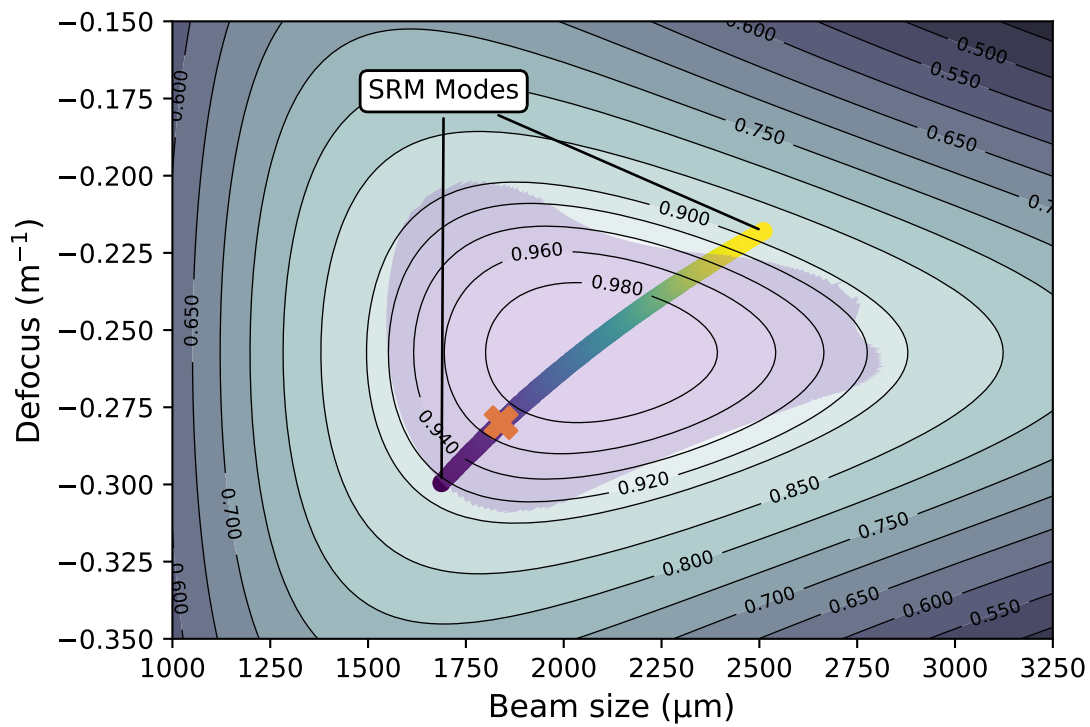


Figure 1.12 (*previous page*): Top panel: the set of modes that could be present at the SRM is shown by the purple-blue-green-yellow curve, as described in chapter ???. The purple region shows which modes can reach 98% mode matching by using the the AWC in HAM6. The orange cross indicates the mode that corresponds to the SRC having a single pass Gouy phase of 21.3° . The contours indicate the mode matching between the arm mode and the OMC mode. Bottom panel: this shows the same modes as the top panel, except they have been transformed to the plane of the OMC waist. The target region is highlighted with a hatched area.

The modes from the SRM that can be matched at to the target is shown in Figure 1.12. The visualise how successful the layout is at the handling likely modes from the arm described in Chapter ??, these modes were compared to the region of WS phase space from which the 98% mode matching requirement from a starting point of 92% mismatch. It was found that almost all likely modes will be covered by the layout, as the modes near the edge of this continuum are more unlikely to occur than the ones near the centre.

For all candidate layouts analysed, there were no major difference in the amount of modes that could be mode matched. Its worth noting that the in figure 1.12, modes at the yellow end of the spectrum may not be physical as they correspond to an unstable SRC. In principle, the requirement for having 45° of Gouy Phase separation between OMx1 and OMx2 doesn't maximise the amount of interferometer modes that can be mode matched. The shape that the purple region will have when HAM6 is constructed will depend on the optical powers of OMx1 and OMx2,

Nevertheless, Figure 1.12 is encouraging since it shows that the actuation provided by the OMx1 and OMx2 is very forgiving, so much so that the strict definition of the role of AWC in HAM6 may be relaxed in order for operators to use the AWC here to correct for other errors in the geometry of LIGO.

1.5 Conclusion about BHD for A+

One of the major changes to LIGO for A+ will be the implementation of BHD. Combined with frequency dependent squeezing, BHD offers the possibility to optimise the detector for different gravitational wave sources. Along with this, BHD

offers a way by which the coupling between the signal recycling cavity length noise and DARM can be reduced.

BHD presents some technical challenges over DC readout. The LO phase noise must be kept low enough as to not contaminate the signal at the AS port with LO path noise. The requirement for the phase noise is shown in figure 1.5. This requirement can be met by using triple suspensions to deliver to local oscillator and AS beams to the beam splitter they are combined on.

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements is shown in figure 1.6. The optical properties of this layout were investigated, and by selection of the right combination of mode matching optics, near orthogonal control of the AS-LO beam alignment and mode matching to the OMC was achieved.

By plotting the $< 98\%$ mode matching requirement in WS space, the robustness to changes in the input beam parameter was investigated. The AWC provides a huge amount of mode matching range that the layout can tolerate very large changes in parameter at the SRM. It was found that almost all possible beams will be able to mode matched to the OMC. This is shown in figure 1.12.

Appendix A

Sensing and Control in the HAM6 Layout

The sensors and actuators for controlling the various degrees of freedom that the HAM6 layout is responsible for is described in brief in the following sections.

A.0.1 Longitudinal Control

Longitudinal control refers to the sensing and actuation of lengths within the interferometer, i.e. position of optics along the optical axis. An example of longitudinal degree of freedom is the differential arm length. There are many longitudinal degrees of freedom within the interferometer and many of them are the lengths of cavities. The degrees of longitudinal degree of freedom that the HAM6 layout must provide sensing and control for are described in the next three sections.

A.0.1.1 Homodyne Angle

One method of controlling the homodyne angle is using the so called ‘RF 36’ signal. LIGO uses 9 MHz and 45 MHz modulations on the light to detect various lengths within the interferometer that need to be controlled. The beat between these two will produce a 36,MHz signal. The homodyne angle determines the amplitude of this signal, and so by demodulating a photodiode measuring the light in reflection from the OMC at 36 MHz the homodyne angle may be controlled. This is detailed in (Tengs Thesis ref). The actuation may be provided by OM0, BHDBS2 or by BHDM1. Alternatively, a dither signal at a few kHz on OM0 could provide can provide a signal corresponding to the homodyne angle. **Cant the homodyne angle be set to 90° by subtracting the photodiode signals and requiring the DC to be zero, as in a Mach zehnder?**

A.0.1.2 OMC Length

Keeping the OMC length locked to the combined LO-AS beam can be done by actuating on the length of PZT mirrors in the OMC. A few kHz signal may be created by the modulation on a HRTS in either the LO or AS path, one of the HDS after BHDBS2, or event a dither on the other PZT within the OMC. These modulations will be measurable in transmission of the OMC. Alternatively, a PDH signal may be created by using the 9 MHz signal may be measured in reflection from the OMC.

A.0.1.3 Squeezing Angle to Measurement Phase

The squeezing and measurement phase may be controlled by measuring the 3,MHz signal transmitted through the OMC. This signal originates in the AOM in the squeezer, and the squeezing angle can be controoled with the PZT in the the squeezer layout in HAM7.

A.0.2 Alignment Control

The alignment of beams within the interferometer is required to maximise the power of the laser the mode containing the gravitational wave signal. The alignment degrees of freedom that need to be sensed and controlled by the HAM6 layout are discussed in the next four sections.

A.0.2.1 LO and AS alignment

The alignment of the signal to the to the AS beam will be done with either OM0 or BHDM1, and BHDBS2. A 36 MHz signal measured in reflection of the OMC can be used to determine the overlap of the two beams. By having a Gouy Phase telescope in reflection of the the OMC, a signal proportional to displacement and angle may be obtained. Alternatively, a dither at a few Hz can be applied to BHDM1 and BHDBS2, and this signal will appear in the light transmitted by the OMC.

A.0.2.2 Alignment of the LO-AS beam to the OMC

A dither applied to a HRTS in either the LO or AS path will produce a signal in transmission through the OMC. The mirrors OMxS and OMx2 will be used to align the LO beam into the OMC, and as these are separated by $\sim 90^\circ$ of Gouy phase,

orthogonal control of the input beam alignment will be achievable.

A.0.2.3 Squeezer to AS beam

The pick-off at OM0 will be used to align the squeezer beam to the AS beam. The squeezer light has 42 MHz sidebands, and so a pair of wavefront sensors separated by $\sim 90^\circ$ of Gouy phase can be demodulated at this frequency to derive an alignment error signal. The squeezer beam will be aligned to the AS beam with optics between the filter cavity and the OFI. The signal and squeezer beam will be aligned onto these wavefront sensors using SR2 and a tip tilt in transmission of OM0.

A.0.2.4 Differential Hard Mode

The differential hard mode arises due to radiation pressure effects in the arm cavities. As the power in the cavity changes, the dynamics of the test mass changes. These dynamics result from the alignment of the cavity axis to the test masses due radiation pressure. This differential hard mode dynamic must be controlled. This signal will be detected with light that is transmitted through OM0, and actuated on with the test masses.

Appendix B

Quantum Treatment for Balanced Homodyne Detection

[12]

Bibliography

- [1] Kiwamu Izumi and Daniel Sigg. Advanced LIGO: length sensing and control in a dual recycled interferometric gravitational wave antenna. *Classical and Quantum Gravity*, 34(1):015001, dec 2016.
- [2] D.E. McClelland, N. Mavalvala, Y. Chen, and R. Schnabel. Advanced interferometry, quantum optics and optomechanics in gravitational wave detectors. *Laser & Photonics Reviews*, 5(5):677–696, mar 2011.
- [3] Haocun Yu, , L. McCuller, M. Tse, N. Kijbunchoo, L. Barsotti, and N. Mavalvala. Quantum correlations between light and the kilogram-mass mirrors of LIGO. *Nature*, 583(7814):43–47, jul 2020.
- [4] The LIGO Scientific Collaboration. A gravitational wave observatory operating beyond the quantum shot-noise limit. *Nature Physics*, 7(12):962–965, sep 2011.
- [5] L. McCuller, C. Whittle, D. Ganapathy, K. Komori, M. Tse, A. Fernandez-Galiana, L. Barsotti, P. Fritschel, M. MacInnis, F. Matichard, K. Mason, N. Mavalvala, R. Mittleman, Haocun Yu, M.E. Zucker, and M. Evans. Frequency-dependent squeezing for advanced LIGO. *Physical Review Letters*, 124(17), apr 2020.

- [6] H. J. Kimble, Yuri Levin, Andrey B. Matsko, Kip S. Thorne, and Sergey P. Vyatchanin. Conversion of conventional gravitational-wave interferometers into quantum nondemolition interferometers by modifying their input and/or output optics. *Physical Review D*, 65(2), dec 2001.
- [7] James Lough, Emil Schreiber, Fabio Bergamin, Hartmut Grote, Moritz Mehmet, Henning Vahlbruch, Christoph Affeldt, Marc Brinkmann, Aparna Bisht, Volker Kringel, Harald Lück, Nikhil Mukund, Séverin Nadji, Borja Sorazu, Kenneth Strain, Michael Weinert, and Karsten Danzmann. First demonstration of 6 db quantum noise reduction in a kilometer scale gravitational wave observatory.
- [8] T. Zhang. *Advanced techniques for future generation gravitational wave detectors*. PhD thesis, University of Glasgow, 2019.
- [9] C. Bond, P. Fulda, A. Freise, D. Brown, K. Kokeyama, L. Carbone, and A. Perreca. Finesse Input Files for the L1 Interferometer. LIGO Document T1300901-v5 <https://dcc.ligo.org/LIGO-T1300901>.
- [10] L. Barsotti, L. McCuller, M. Evans, and P. Fritschel. The A+ design curve. LIGO DCC, April 2018. LIGO Document T1800042-v5 <https://dcc.ligo.org/LIGO-T1800042/public>.
- [11] B. Lantz, R. Mittleman, J. Kissel, J. Warner, and A. Pele. Typical Motion of HAM Suspension Points in O2. LIGO Document T1800066-v2 <https://dcc.ligo.org/T1800066>.
- [12] Sebastian Steinlechner, Bryan W. Barr, Angus S. Bell, Stefan L. Danilishin, Andreas Gläfke, Christian Gräf, Jan-Simon Hennig, E. Alasdair Houston,

Sabina H. Huttner, Sean S. Leavey, Daniela Pascucci, Borja Sorazu, Andrew Spencer, Kenneth A. Strain, Jennifer Wright, and Stefan Hild. Local-oscillator noise coupling in balanced homodyne readout for advanced gravitational wave detectors. *Physical Review D*, 92(7), oct 2015.